

## EDUCATION

• **Booth School of Business, University of Chicago, IL** **September 2026-Present**

*PhD in Economics*

• **Michigan State University, East Lansing, MI** **GRE: 170Q/166V** **May 2024**

*Honors Dual Degree in Economics and Molecular Genetics & Genomics, Minors in Mathematics and Law & Public Policy*

### • **Additional Academic Achievements**

*NSF Graduate Research Fellow:* for scholars with demonstrated potential to make significant contributions to research

*Doctoral Centennial Fellow:* for exceptional promise as a future scholars, awarded by the Booth Economics Department

*George F. McGregor Scholar:* for econometric ability, awarded by Professor Wooldridge and Professor Soren Anderson

*First Place UURAF Poster Presentation:* for Sexual Dimorphism in Lupus

*MSU Economics PURI 2022 Grant Winner:* provided top research proposals with 1 year of funding

*MSU Econ Scholars Program:* recognizing top economics undergraduates

*Dean's List every semester in the college of Natural Science and the college of Social Science*

*Hanna M. Rasmussen Memorial Alumni Distinguished Scholar:* full-ride merit-based scholarship to Michigan State University

## RESEARCH EXPERIENCE

• **Stores Economics and Applied Science (SEAS) at Amazon** *June 2024-Present*

*Economist Apprentice*

Amazon, Seattle, WA

– Research with Dr. Nichols, Dr. Jahedi, Professor Imbens, and Professor Card on network externalities, labor productivity, worker health, consumer spending, and high-dimensional causal inference methods.

• **Liang Laboratory** *September 2022-2025*

*Professorial Assistantship*

Michigan State University, East Lansing, MI

– Second author for a paper on sex-linked lupus bias in XIST and RPII KD fetal mouse keratinocytes–2025 publication

• **MSU Department of Economics** *September 2022-2023*

*Provost's Undergraduate Research Initiative Researcher*

Michigan State University, East Lansing, MI

– Research on incorporation of Kahneman and Tversky prospect theory for a model of symmetric information with a sender-specific signal for a receiver taking a non-contractible action affecting the welfare of both players.

• **MSU Department of Mathematics** *September 2022-2023*

*Honors Research Project*

Michigan State University, East Lansing, MI

– Honors Project on Graph Theory and properties of Cayley graphs in Abstract Algebra and Number Theory

• **Robison Laboratory** *September 2021-2022*

*Professorial Assistantship*

Michigan State University, East Lansing, MI

– Received an NSP grant for analysis of hypothalamic cell deviation, drug dependency, cPTSD, and reward-seeking behavior

• **Pyeon Laboratory** *September 2020-2021*

*Professorial Assistantship*

Michigan State University, East Lansing, MI

– Protein assays and genetic analysis around chemokines associated with HIV and cervical cancer in human cell lines.

• **Nobrega Laboratory** *June-August 2019*

*Research Internship*

Human Genetics Department, University of Chicago, Chicago, IL

– Research with Dr. Nobrega examining the role of epigenetics on obesity, analyzing RNA sequencing data using R.

## WORK EXPERIENCE

• **MSU Department of Economics** *September 2022-May 2024*

*Undergraduate Social Media Coordinator*

East Lansing, MI

– Management of the Department's social media pages—posting relevant economics literature/departamental content

• **MSU College of Social Science** *November 2023-May 2024*

*Pathway to Research Program Student Mentor*

East Lansing, MI

– Mentor for first-generation and diverse social science students—helping them engage in research, find fields and mentors in which they're interested, communicate with potential mentors, and secure a research position.

• **Oliver Wyman** *July 2023-August 2023*

*Core Consulting Group Intern*

New York City, NY

– Created spend models, slide decks, and market evaluations of metals for an aerospace engineering merger.

– Tutor/mentor for students entering consulting, focusing on networking for diverse talent with MSU SCA

• **Johnsen Fretty and Co** *May 2023-July 2023*

*Mergers and Acquisitions Intern*

Stamford, CT

– Evaluated value and purchasing models of potential acquisitions, particularly in adjustments for COVID impact.

• **MSU Department of Computational Mathematics, Science, and Engineering** *December 2021-May 2022*

*Undergraduate Teaching Assistantship*

East Lansing, MI

– Assistantship geared toward undergraduate tutoring of introductory theoretical statistics and probability.

TECHNICAL SKILLS AND INTERESTS

**Languages:** Fluent in Hindi, Marwari, and Gujarati and proficient in Latin and Spanish  
**Coding Languages:** Skilled in Python, R, SQL, Spark, and STATA  
**Markup/Document/Replication Tools:** Skilled in LaTeX, markdown, Jupyter NB, VSCode, Git, & Github  
**Coursework:** Behavioral Economics, Advanced Econometrics, Differential Equations, Linear Algebra, Abstract Algebra, Real Analysis(Harvard Extension School), Economics of Labor & Markets, Theory of Probability & Theory of Statistics

POSITIONS OF LEADERSHIP

•**Undergraduate Liaison,** MSU Economics Department’s DEI Initiative Committee *January 2022-May 2024*  
•**Vice President,** Women and Minorities in Economics (WAMIE) *January 2022-May 2024*

RESEARCH PAPERS/PRESENTATIONS

**The Impact of Abortion Bans After Dobbs v. Jackson on Teen Birthrates and Education Attainment:** Working Paper  
We find statistically significant evidence that abortion bans increase teen birth rates in the United States, using Wooldridge’s new ETWFE specification to address heterogeneous pre-trends. However, concurrent COVID-19 policies complicate the analysis of abortion bans’ effects on educational attainment, as states with bans also implemented less stringent pandemic restrictions, requiring further research to isolate causal effects.

**Bayesian Persuasion and More: Applying Prospect Theory to Information Signaling Games:** Symposium poster  
This research investigates how incorporating Kahneman and Tversky’s prospect theory into the Kamenica and Gentzkow Bayesian Persuasion model affects sender-receiver signaling games, hypothesizing that probability distortion breaks down the original model’s predictions in single-play scenarios while maintaining similar aggregate results. Through analysis of probability distortion and diminishing sensitivity in a prosecutor-judge framework, we demonstrate that prospect theory provides a theoretical explanation for why Bayesian Persuasion rarely occurs in real-world environments, as distorted probability perceptions can fundamentally alter receiver decision-making.

**TNF $\alpha$  Exposure in XIST and RPII KD Lupus-Prone Mice Stimulates Sex-Bias:** Manuscript to be published in 2025  
We investigate the role of XIST RNA in inflammation and lupus pathogenesis, as sexual dimorphism in immune regulation results in women suffering 8 out of 9 lupus cases. Through qPCR analysis and immunohistochemistry of female mouse keratinocytes with XIST knockdown treated with TNF- $\alpha$  inflammatory factor, we identified significant differential expression of inflammation-related genes (including NF $\kappa$ BIA, TNFSF11, and NF- $\kappa$ B pathway components), preliminarily mapping an epigenetic relationship between XIST RNA and altered inflammatory protein expression that may explain sex-based differences in autoimmune disease incidence.

**Nesterov’s Implicit Regularization Under Stochastic Gradient Descent:** Working Paper  
We study Nesterov’s Accelerated Gradient (NAG), finding that it exhibits stronger implicit regularization than standard Gradient Descent and Polyak’s Momentum, with the regularization strength increasing by factors of  $((\frac{1+\beta}{1-\beta})^2)$  and  $(1 + \beta)$  respectively. Through theoretical analysis using existing ODE methods to calculate error bounds and experimental validation, we show that higher momentum parameters in NAG reduce variation and enhance test accuracy by inducing a differentiable trajectory with a more powerful implicit regularizer.

**Skippy FC-FC Cost Curve Live Experiment Evaluation:** Manuscript submitted to the Amazon Machine Learning Conference (AMLC)  
Amazon Confidential.

**Amazon City:** Best Economics Paper at Amazon-internal Consumer Science Summit (Last Author)  
Amazon Confidential

**Mentoring Matters: Ambassador Quality and New Hire Productivity in Fulfillment Centers:** Working Paper  
Amazon Confidential